

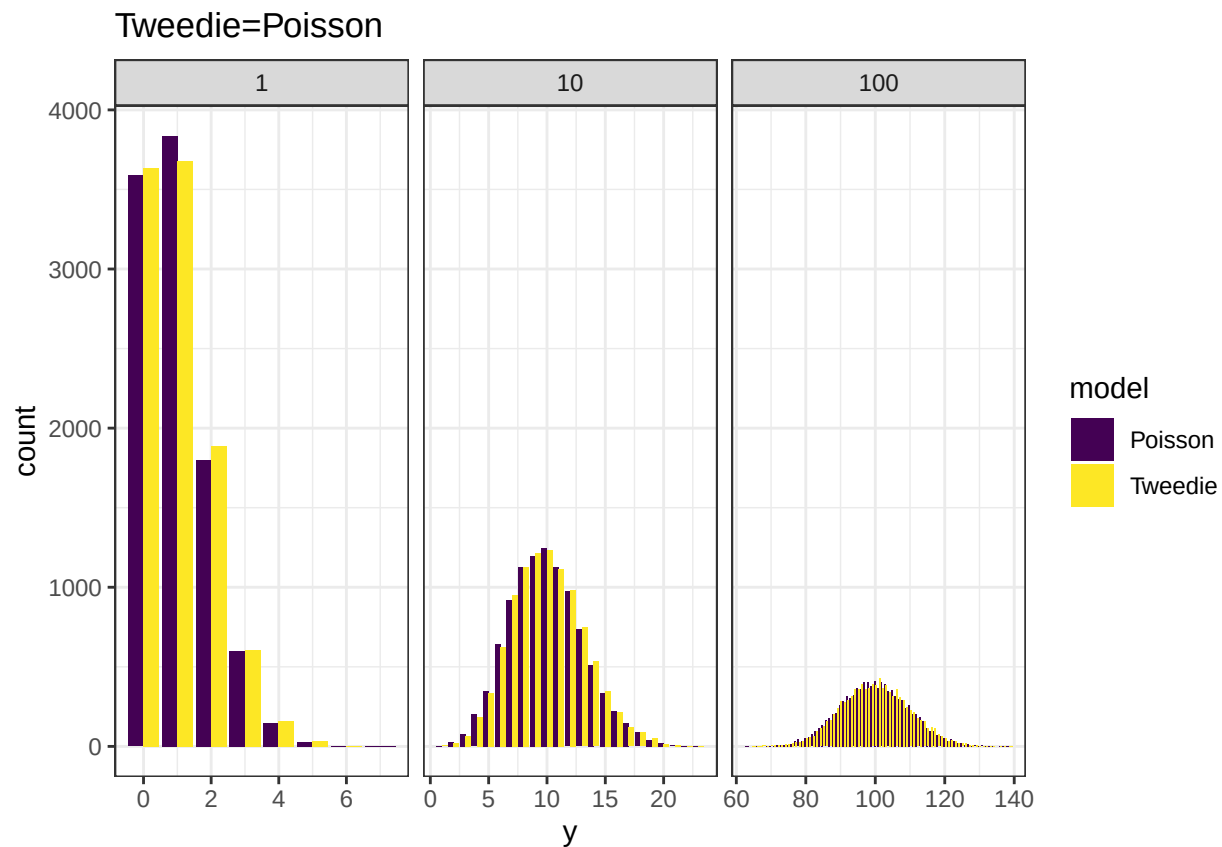
Catch, effort and CPUE

Beth Babcock

2025-11-03

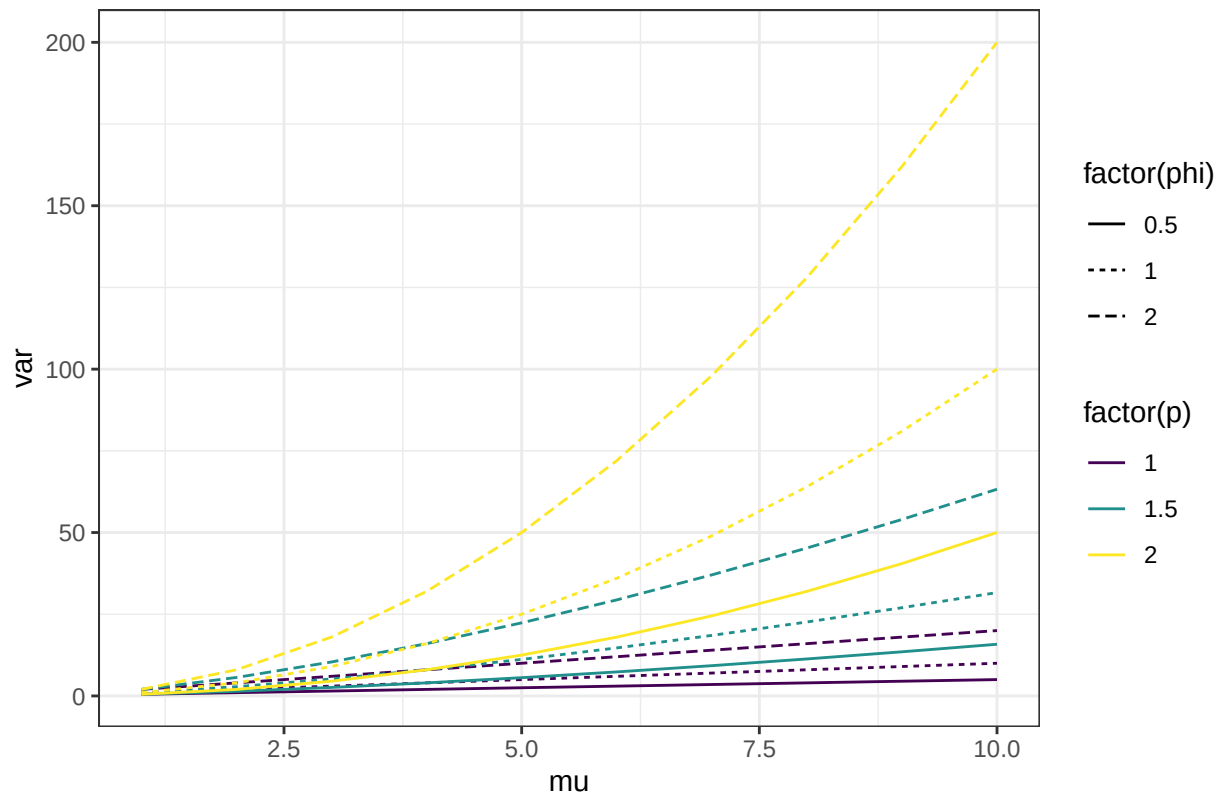
```
library(tidyverse)
library(tweedie)
theme_set(theme_bw())
update_theme(palette.colour.discrete = scales::pal_viridis())
#Tweedie=poisson
df<-expand_grid(mu=c(1,10,100),sim=1:10000,p=1,
                phi=1,model=c("Poisson","Tweedie")) %>%
  rowwise()%>%
  mutate(y=ifelse(model=="Tweedie",rtweedie(1,mu=mu,phi=phi,power=p),
                  rpois(1,mu)))

ggplot(df,aes(x=y,fill=model))+
  geom_bar(position = position_dodge())+
  facet_grid(cols=vars(mu),scales="free")+
  ggtitle("Tweedie=Poisson")
```



```
#Tweedie variance
df<-expand.grid(mu=seq(1:10),p=c(1,1.5,2),
                phi=c(0.5,1,2),model=c("Tweedie")) %>%
  mutate(var=phi*mu^p)
ggplot(df,aes(x=mu,y=var,color=factor(p),linetype=factor(phi)))+
  geom_line()+
  ggtitle("Variance with mean, p, and phi")
```

Variance with mean, p , and ϕ



```
#Tweedie with zeros
df<-expand.grid(mu=c(1,10),sim=1:10000,p=1.15,
               phi=c(1,2),model=c("Tweedie")) %>%
  rowwise()%>%
  mutate(y=rtweedie(1,mu=mu,phi=phi,power=p))
ggplot(df,aes(x=y,fill=factor(y==0)))+
  geom_histogram()+
  facet_grid(cols=vars(mu),rows=vars(phi),scales="free")+
  ggtitle("Tweedie with zero inflation")
```

Tweedie with zero inflation

